

Elliott Sound Products

Sinewave Oscillators

Sinewave Oscillators - Characteristics, Topologies and Examples

© 2010, Rod Elliott (ESP)
Updated May 2024


[Main Index](#)

[Articles Index](#)

Contents

- [Introduction](#)
- [1 - Digital Synthesis](#)
 - [1.1 - Adding Filters](#)
- [2 - Essentials Of An Oscillator](#)
- [3A - The Wien Bridge Oscillator](#)
- [3B - Two Opamp Wien Bridge](#)
- [4 - Amplitude Stabilisation](#)
 - [4.1 - Thermistor Stabilisation](#)
 - [4.1.1 - The Venerable \(And Unobtainable\) R53](#)
 - [4.2 - Lamp Stabilisation Details](#)
 - [4.3 - LED/ LDR Stabilisation](#)
 - [4.4 - Electronic Stabilisation](#)
- [5 - Wien Bridge Alternatives](#)
 - [5.1 - State Variable Filter](#)
 - [5.2 - Quadrature Oscillator](#)
 - [5.3 - Oscillating Filter](#)
 - [5.4 - Low-Pass Filter + Integrator](#)
 - [5.5 - Phase-Shift + Distortion Cancellation](#)
 - [5.6 - Phase Shift Oscillators](#)
 - [5.6.1 - Buffered Phase Shift & 'Bubba' Oscillators](#)
 - [5.7 - BFO - Beat Frequency Oscillator](#)
 - [5.8 - Squarewave Oscillator + Filter](#)
 - [5.9 - Twin-T Oscillator](#)
- [6 - L/C Oscillators](#)
- [7 - Waveform Shaping](#)
- [8 - Digital Generation](#)
- [9 - Tuning Element & Component Selection](#)
- [10 - Spot Frequencies](#)
- [11 - Distortion And Level Stability.](#)
- [Conclusion](#)
- [References](#)

Introduction

Audio oscillators (aka audio signal generators) have been an essential piece of test gear for many decades. While laboratory instruments were available (at laboratory prices) from the very early days, it wasn't until the late 1950s that affordable signal generators became available for hobbyists. Most were still frighteningly expensive and of mediocre performance by modern standards, but there has been a ready supply of such instruments for



professionals and hobbyists alike for many years now. Kit versions have been made by many different companies and 'hobby' circuits have been published in electronics magazines for as long as I can remember.

An intriguing conundrum on the Net is the constant belly-aching from many vested interests that "sinewaves are simple", and are therefore a poor test of an amplifier's distortion performance. If this were true, then a low distortion sinewave oscillator would not pose any problems to build, indeed, it too would be 'simple'. This being the case, I challenge those who believe this nonsense to build a simple variable frequency sinewave generator with minimal (or no) distortion. It's simple, isn't it?

Alas, this is not the case, and there are many different schemes published that desperately attempt to obtain a low distortion sinewave, without having to revert to complex high-bitrate digital synthesis, and without using the venerable (and now unobtainable) R53/ RA53 or similar NTC thermistor. Even a sinewave generator that has low distortion at one or two spot frequencies isn't easy, and a variable generator takes the difficulty to another level. Ideally, a high purity sinewave generator will *not* require tuned filters at the output to reduce distortion.

In this article, I will concentrate on variable frequency oscillators, because while spot frequencies can be useful if you only need to check distortion at a couple of frequencies, most people like to be able to test filters, amplifiers, loudspeakers, and other devices that are generally expected to be able to reproduce more than one (or two) frequencies. However, there are many requirements for single-frequency sinewave oscillators, so they are not avoided altogether.

The distortion should ideally be as low as possible, but anything below 0.1% starts to become rather difficult with many of the methods available. It's certainly possible to improve on this, but very careful adjustment of all the parameters (time constants, allowable stabilisation range, etc.) is needed to get good results. Some comparatively simple arrangements can give very good results, but only over a limited range (using common and readily available opamps). Indeed, opamps impose many additional limitations. Distortion is usually well within acceptable limits, but not many low-cost opamps will allow operation of any oscillator topology to much beyond ~30kHz. This is very limiting, as it is common for general purpose audio oscillators to have a range up to at least 100kHz, preferably more.

Note that this article is very specific - it deals only with 'linear' oscillators - those that are designed to generate a sinewave. Even more specifically, the range is limited to audio frequencies, plus at least a couple of octaves either side. Most audio oscillators are expected to be able to cover the range from about 5Hz up to at least 100kHz.

You won't find any multivibrators or other square/rectangle generators here, nor will you find RF oscillators, other than by a glancing reference.

Note that the schematics presented here are for the purpose of illustration and education, and should not be considered to be fully functional as shown. In many cases the circuits will work as described, but this is not guaranteed and cannot be assumed. Some circuits incorporating feedback stabilisation loops using other than lamps or thermistors may require some effort to ensure stability under actual operating conditions. This is an article that describes the principles - it is not a collection of projects that have been built and fully debugged.

No descriptions are provided for common function generator ICs. Basic function generators have been around for some time now, and there are several specialised ICs designed for just that purpose. However, most have mediocre distortion performance (typically around 1% for the better versions), and that limits their usefulness. Some of the ICs include the Exar XR2206, Maxim MAX038 and Intersil ICL8038, but not all are still available because they are now obsolete. If you are interested, look up the details for them - provided you don't need low distortion, one of them may be just what you need. Most use 'waveform shaping' to get a passable facsimile of a sinewave (see [Section 7 - Waveform Shaping](#) for an example).



Several low cost function generators are available on-line, and most use one of the common ICs. You'll also see many DDS function generators at fairly low cost. None of the cheap function generators are suitable if low distortion is needed.

Note that power supplies and bypass capacitors are not shown in the drawings that follow. Most of the opamp circuits in this article will require ± 12 -15V DC supplies. Although all gain blocks are shown as opamps, in many cases you will have to build a discrete 'opamp' or the circuit will not be satisfactory at high frequencies. Most IC opamps will typically be ok up to perhaps 30kHz or so, but if you need good performance up to 100kHz or more, you'll almost always need a discrete circuit or a *very* fast opamp. All opamps (whether IC or discrete) need ceramic power supply bypass caps close to the IC or other parts to prevent instability (either parasitic or continuous RF oscillation). Discrete opamps may be needed if a lamp is used for stabilisation, because the current needed is high enough to cause many opamps to increase their distortion beyond the quoted figures.

An oscillator has two very specific requirements. The amplifier must be able to perfectly match the losses in the frequency determining network. The frequency determining network must be arranged so that the signal fed back to the amplifier results in *positive* feedback. If the gain exceeds that required for oscillation, the output will increase until it's distorted, and if too low the oscillation will die away to nothing. These constraints apply to all analogue oscillators.

By definition, oscillators do not require an externally applied input signal, but instead use part of the output signal via a frequency selective feedback network as the input signal. It is the circuit noise and/ or offset voltage that provides the initial 'trigger' signal to the circuit when positive feedback is employed. If the gain criterion is satisfied, the output builds up over a period of time, oscillating at the frequency set by the circuit components ^[9].



Note: The capacitors in frequency networks will typically be MKP (polypropylene) caps for high performance, or MKT (polyester) for a general purpose unit. 1% metal film resistors are recommended in all cases. Polypropylene is probably one of the better options where you need high stability. *Never* use multilayer ceramic caps in any oscillator unless you actually want it to have high distortion and very poor (and unpredictable) frequency stability with temperature variations (this isn't a common requirement in my experience). 😊 Polyester (PET, Mylar, etc.) caps have a positive temperature coefficient, and polypropylene is negative (but smaller). Polystyrene caps are very good in this role, but they are hard to get and are only available in fairly low values.

There is a very low distortion sinewave generator published as [Project 174](#) that you may find useful if you need to build an audio oscillator. It uses a novel sample-and-hold circuit to achieve amplitude stabilisation, and distortion+noise is around 0.001% with good opamps. The oscillator circuit is the same as that shown in Figure 5, and amplitude control uses an LED/ LDR optocoupler. Another is [Project 179](#), which uses a discrete circuit and a lamp with a 'padding' network to minimise the lamp induced distortion (particularly at low frequencies).

Many early sinewave oscillators used a dual variable capacitor for tuning. This is a good option, but it means that all resistor values are high to very high, which affects noise performance and makes the circuitry susceptible to stray capacitance. This option has not been included in any of the circuits shown, but can be used if you have a suitable dual tuning capacitor available. A significant advantage is that variable capacitors tend not to become noisy like potentiometers, and their tracking is usually better. The minimises amplitude 'bounce' as the frequency is changed. Given that you'll be hard-pressed to find a tuning gang of more than 500pF or so, that means resistors have to be 20M Ω just to get down to 16Hz. Such high values also increase thermal noise from the resistors themselves, but this isn't as much of a problem as you may expect.



1 - Digital Synthesis

I mention this first because many people will be tempted by cheap DDS modules that are available from many on-line suppliers. These look like a good idea at first, but you'll almost certainly become annoyed rather quickly because they are nowhere near as good as the specifications *seem* to indicate, and they generally have fairly poor distortion performance. An analogue circuit (and interface) will always be easier to use. Frequency and level accuracy of analogue circuits may appear poor, but mostly we don't care too much about absolute accuracy.

Many of the latest and greatest oscillators use DDS - direct digital synthesis, but such units are usually quite expensive. There are some very cheap ones, but they are not suitable for serious measurements. Even a 12-bit output is barely acceptable, as this will cause the minimum distortion to be around 0.04% - not bad, but certainly not very impressive. Lower resolution means higher distortion - 8 bit resolution gives a theoretical 0.5% THD with basic sample-rate filtering, and anything less than 8 bits is obviously pointless. While this can be improved with more advanced filtering, this increases complexity. For a usable system, I would not be happy with anything less than 14 bits, and preferably 20 bits or more. Needless to say, the digital clock frequency needs to be far greater than the highest output frequency. Distortion of a digitally generated sinewave with only sample-rate filtering falls by 6dB for each additional bit (the distortion is halved), so if we start from 7 bits (1%), 8 bits is 0.5%, 10 bits is 0.125%, etc.

Unlike a digital audio format, very steep low pass filters (to remove the switching waveform) usually cannot be used with test equipment. This isn't because the filters are audible or create problems as such, but because the filters need to track the audio signal across the wide frequency range generally available - typically from less than 0.1Hz up to 5MHz or more. Tracking filters expected to cover that range are not easy to implement.

The performance of test equipment should generally be at least 5-10 times better than the device under test ('DUT'). If this is not the case, you can't measure the response of an amplifier accurately if its response approaches that of the measurement system (both input and output devices). Needless to say, measuring an amplifier's distortion using a source that has perhaps 5 to 10 times *more* distortion than the amplifier is a completely pointless exercise. Even if the distortion is the same for the source and the DUT, the reading you obtain is obviously inaccurate and cannot be used meaningfully. This is a constant problem with most workshop systems - even those that are comparatively advanced.

While 'DDS' might be the current buzzword for audio generation, and offer many additional features that most users will probably never use. It's important to understand the limitations of any test equipment that is driven via a menu, push-buttons (or a keyboard) and expects you to read an LCD to see the current settings. Compare this with an 'analogue GUI' where the knob pointer shows the setting, and you can just turn a knob to increase/ decrease amplitude, frequency or range. One small movement vs. many button-pushes wins every time.

For a number of reasons, I mainly use a digital waveform generator these days, but it's actually a pain the bum compared to a fully analogue version. Now that I don't need the very low frequency ability of the digital generator so often, I will be changing back (after some essential repairs, since my preferred unit is now almost 50 years old).

While many people expect that 'newer is better', that is most certainly not the case if functionality is sacrificed for bells and whistles. I do like the ability to press a button and get tone-bursts, but I don't like changing from sine to squarewave output, and having the oscillator spontaneously reset my sinewave level (yes, it does that, and it's bloody annoying!). I have no idea why anyone thought that was 'useful'.



1.1 - Adding Filters

One thing that's fairly recent in my workshop/ lab is the addition of a high-Q tuned filter. My most often used distortion meter is a fixed frequency unit, and operates at 400Hz and 1kHz. By adding a pair of filters with one tuned for each frequency of interest, I now have measurement capability that's around 0.007%, limited by the distortion meter itself. The filters I used are shown in the article [Gyrator Filters, Figure 24](#), and I can simply switch from one to the other as needed. These filters have reduced the distortion from my arbitrary waveform generator from 0.02% to well below the resolution of the distortion meter. Analysis of distortion is *always* accompanied by monitoring the distortion meter's residual output, as that is crucial to understand the composition of the distortion. If it shows sharp spikes or significant excess noise, I know what to look for. This is an important step, but it's not done often enough so the true nature of the distortion components is hidden, with only a percentage THD provided.

Providing only a distortion percentage can hide some very unpleasant surprises, and I've been monitoring the residual for as long as I've been taking distortion measurements. Most (but sadly, not all) distortion meters have an output for just this purpose, and observing the results on a scope or listening to the residual on a monitor speaker can tell you a great deal about the exact nature of the distortion 'artifacts' produced by the device under test.

2 - Essentials Of An Oscillator

When we speak of audio oscillators, the primary waveform is a sinewave. Having access to a squarewave is useful, but the sinewave is favoured for the vast majority of tests. If we wish to measure distortion, then the sinewave needs to be exceptionally pure, with a THD that is substantially lower than that of the device under test. While less than 0.01% THD is desirable, it is extremely difficult to achieve with any variable frequency oscillator. Obtaining very low distortion is comparatively easy for a single frequency tone generator, but these are not common because few people can afford the space or cost of a dedicated oscillator that can't also be used for general purpose tests.

Most oscillators are simply an amplifier, with a tuned circuit (frequency selective filter) of some kind to set the frequency. In order to oscillate, it requires positive feedback. The amount of positive feedback needed is determined by many factors, including the losses through the selective filter. It is the filter that determines the frequency, and it can be either an all-pass (phase shift) or band-pass type. Band-pass filter based oscillators have a theoretical advantage, in that any distortion created by the amplitude stabilisation network is subjected to the action of the filter, so in theory distortion should be lower. In reality, this is not necessarily the case.

In the tests I did for this article, I found that the filter doesn't make as much difference as one might expect. Even though the Wien bridge (the most commonly used audio oscillator topology of all) has only very basic filtering, it still has amongst the lowest distortion of any of the different types. The Wien bridge is common for a number of reasons, not least being that it has good frequency stability, is a simple circuit, and is easily tuned over a one decade (10:1) range. The general schematic of a more or less typical Wien bridge oscillator (one of the most common types) is shown below. We will then dissect the various parts so that operation is easily understood.

As will be shown later, there are many different schemes for oscillators. Some are good, and others less so. For acceptable distortion, very few diode or zener stabilised oscillators are suitable, however there is one exception that will also be discussed. Almost any clipping stabilisation scheme can be replaced with a thermistor (best), an LED/LDR opto coupler (good) or a junction FET (varies from useless to good). Unfortunately, as we have already seen, thermistors that are usable for this application are virtually impossible to obtain. Occasionally R53/RA53 thermistors appear on on-line auction sites, but these are a rather unreliable source at the best of times.



considering that the amplitude is limited only by the diodes in the NIC circuit. Output level is around 620mV peak, or 440mV RMS. The impedance of the NIC is nominally $-30k$, set by R1. This particular NIC circuit is one of the most common you'll come across, but this is the first time I've seen it used as anything other than a curiosity. The two diodes limit the negative impedance region and thus the amplitude of the oscillation.

It works because a negative impedance is inherently unstable, and when connected to a tuned circuit (C1 plus the gyrator inductor) it will oscillate. Many years ago, tunnel diodes (which have a negative impedance region) were common for RF oscillators, requiring only the diode, tuned circuit and a low voltage power source. While tunnel diodes were moderately common in the 1960s and 70s, they are much less so today. If you want to know more about them, a web search will provide many circuits and explanations.

This is an interesting circuit, and readers may well find it to be useful. It uses two rather 'off-the-wall' concepts, the [NIC](#) and the [gyrator](#). These are both fascinating and obscure, although the linked articles on the ESP site are dedicated to the two circuit ideas. The 'common' NIC shown above is seen all over the Net as an example of how to build a NIC, but working examples are hard to come by.

The circuit shown is a simplified version of one sent to me by Steven Dunbar, AD0DT.

7 - Waveform Shaping

Diode waveform shaping is very common with low cost analogue function generators (it's inside the IC itself in most cases), but the lowest distortion that is typically available is around 1%, although as low as 0.25% is claimed in some literature. The input signal is a triangle waveform, and the diodes progressively clip the peaks to give a reasonably smooth sinewave. Although the distortion is usually audible, it is still usable for simple signal sweeps, for example to find the resonant frequency of a speaker. There are countless different diode clipping schemes on the Net, and the one shown below is purely an example.

Triangle waveforms are very easy to generate with simple opamp circuits, and that makes it attractive for low-cost function generators.

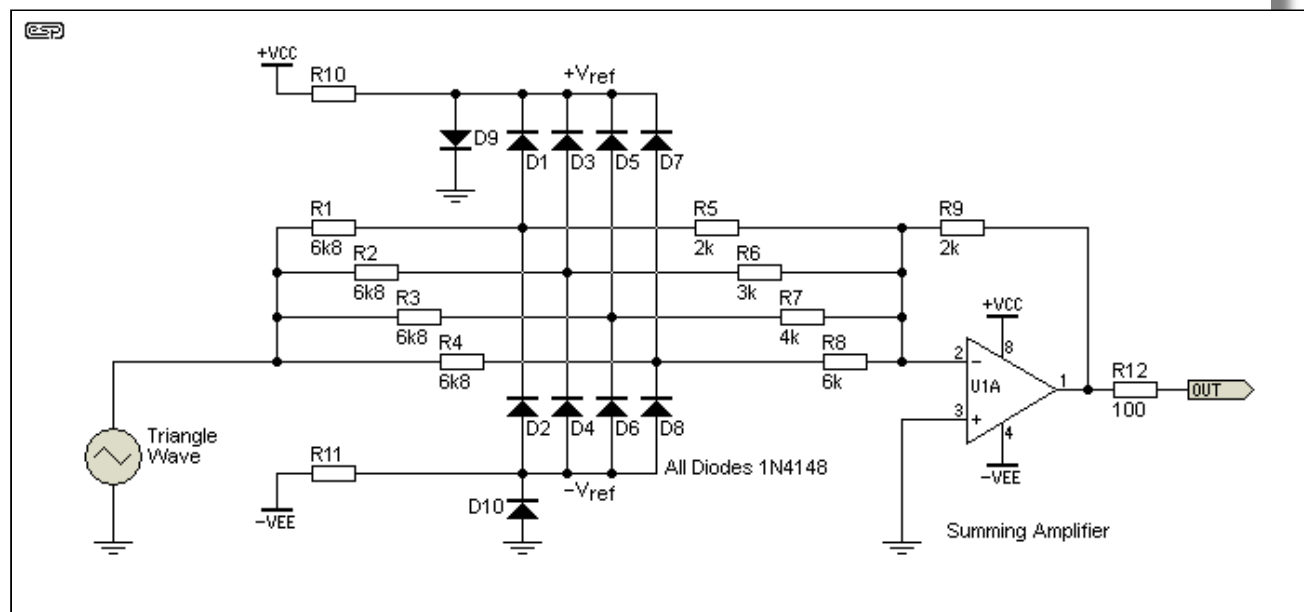


Figure 7.1 - Waveform Shaping Example

In the example shown, a $\pm 6.6V$ triangle wave input gives the lowest distortion. Because of the different impedances in each of the 4 clipping circuits, the output amplifier sums a variety of clipped and un-clipped waveforms, with the end result looking rather like a sinewave. It is very important that the triangle waveform is perfectly symmetrical, or even-order harmonic distortion rises rapidly.



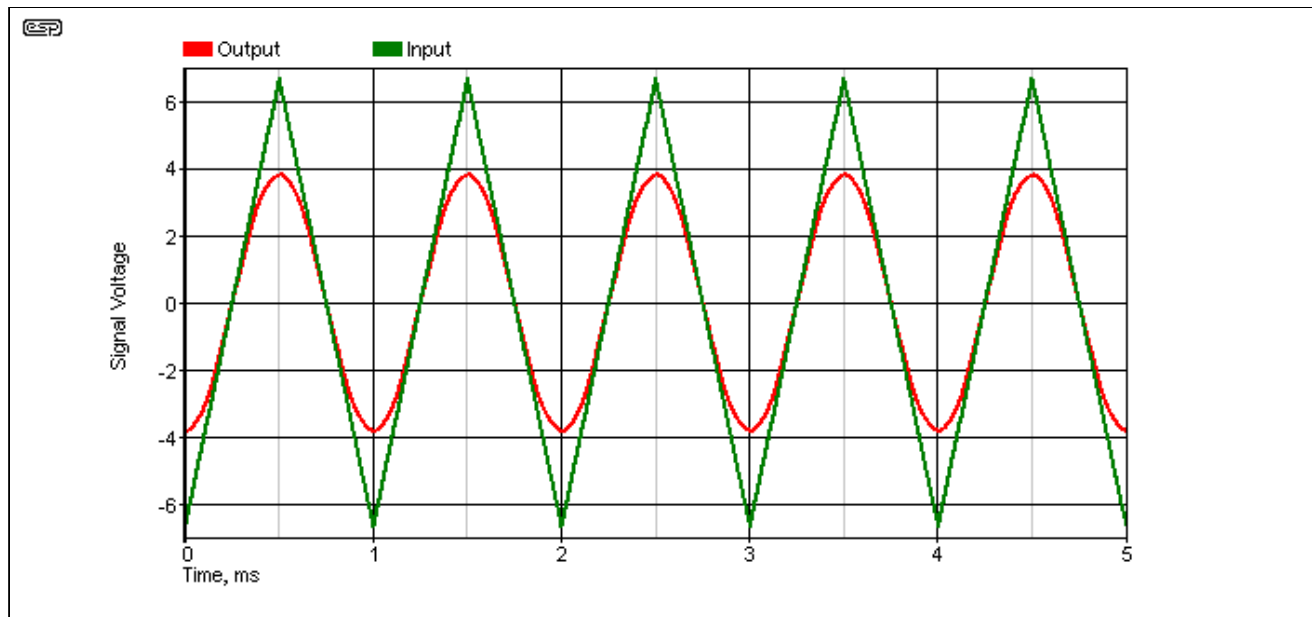


Figure 7.2 - Waveform Shaping Input And Output

The distortion on the output is about 1%, which is fine for basic tests, but is obviously useless for measuring distortion. It's generally not mentioned, but the output waveform will typically have more higher order harmonics than low order. For example, a FFT plot of the output shows a little 3rd harmonic (at about -58dB), but over 10dB more 5th (-42dB), 7th (-46dB) and even 6dB more of the 9th (-51dB). A basic filter can get THD below 1%, but it's hard to improve on that without additional complexity. This method is suitable for integration, but as a discrete circuit there's too many parts for a rather poor end result.

Note that the output was inverted so the direct relationship can be seen (the circuit shown inverts the output signal).

Another option is to use a logarithmic amplifier. While this is theoretically better than diode clipping, in reality there's usually very little difference between them. Unless proper temperature controlled log ICs are used there will usually be a small change in amplitude and distortion as the ambient temperature changes - this applies to diode shaping as well, although the effects are likely to be less severe with the simpler diode clipping circuits.

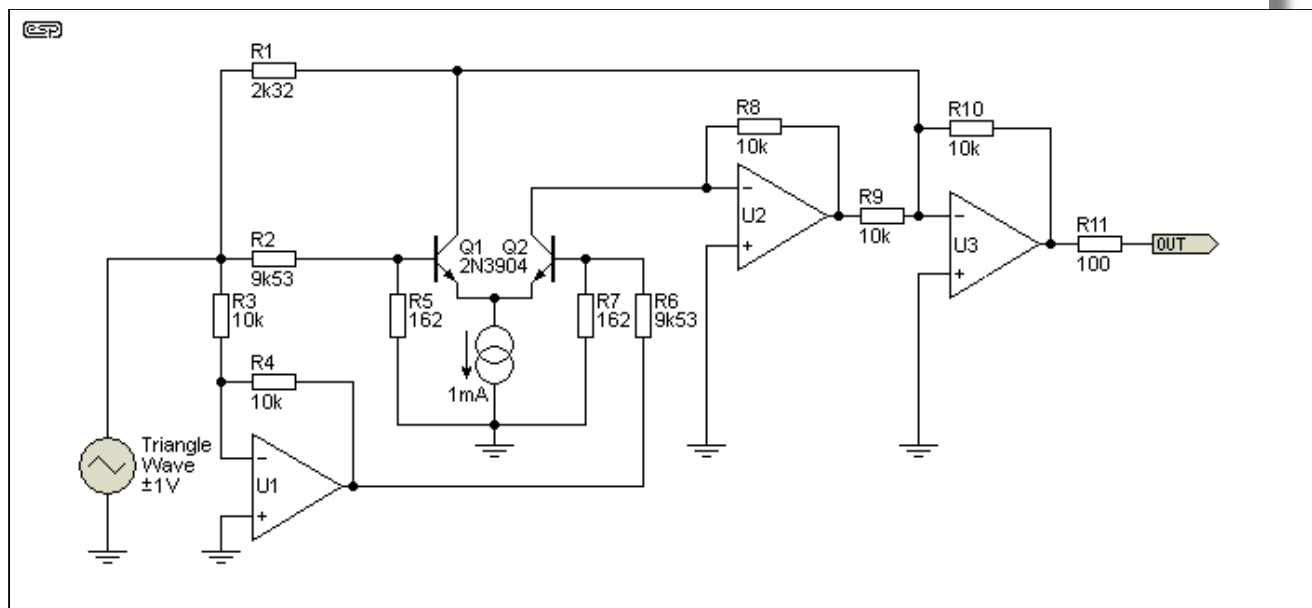


Figure 7.3 - Improved Sine Approximation



By using a transistor differential pair, it's possible to synthesise a triangle wave into a respectable approximation of a sinewave ^[13]. The two transistors must be matched and in close thermal contact with each other. The circuit's been simplified, showing a current source for the emitters, but this needs to be made using additional transistors. Power to the opamps is not shown. Note that the output impedance of the triangle wave generator must be as low as possible (an opamp buffer is recommended).

The circuit as shown is capable of less than 0.2% distortion, which is considerably better than that obtained using the diode clipping circuit shown in Figure 7.2. The reference document contains the formulae for calculating the values, which are quite critical. The amplitude of the triangle waveform must be exactly $\pm 1\text{V}$ for best performance. A deviation of only 100mV causes the distortion to rise dramatically. The requirement for odd value resistors (E48 Series, 48 values per decade) is limiting, and most hobbyists won't have them in stock. Most hobby suppliers only stock the E24 Series, and some don't even go that far.

8 - Digital Generation

Digitally generated sinewaves are becoming much more common than they once were, but most have a limited bit depth which limits the usefulness of such techniques. I would suggest that anything less than 8 bits is completely useless, because distortion will be too high. Eight bit resolution gives a theoretical distortion of 0.5%, and this is halved for each additional bit used. Two techniques that reduce distortion are the addition of a filter (preferably tracking) to remove the harmonics, and adding dither - essentially random (white) noise - at a very low level. For serious work, nothing less than 14 bits is really much use, as distortion is still too high unless post-filtering and dithering is used. A 14 bit system should be able to provide distortion below 0.01%.

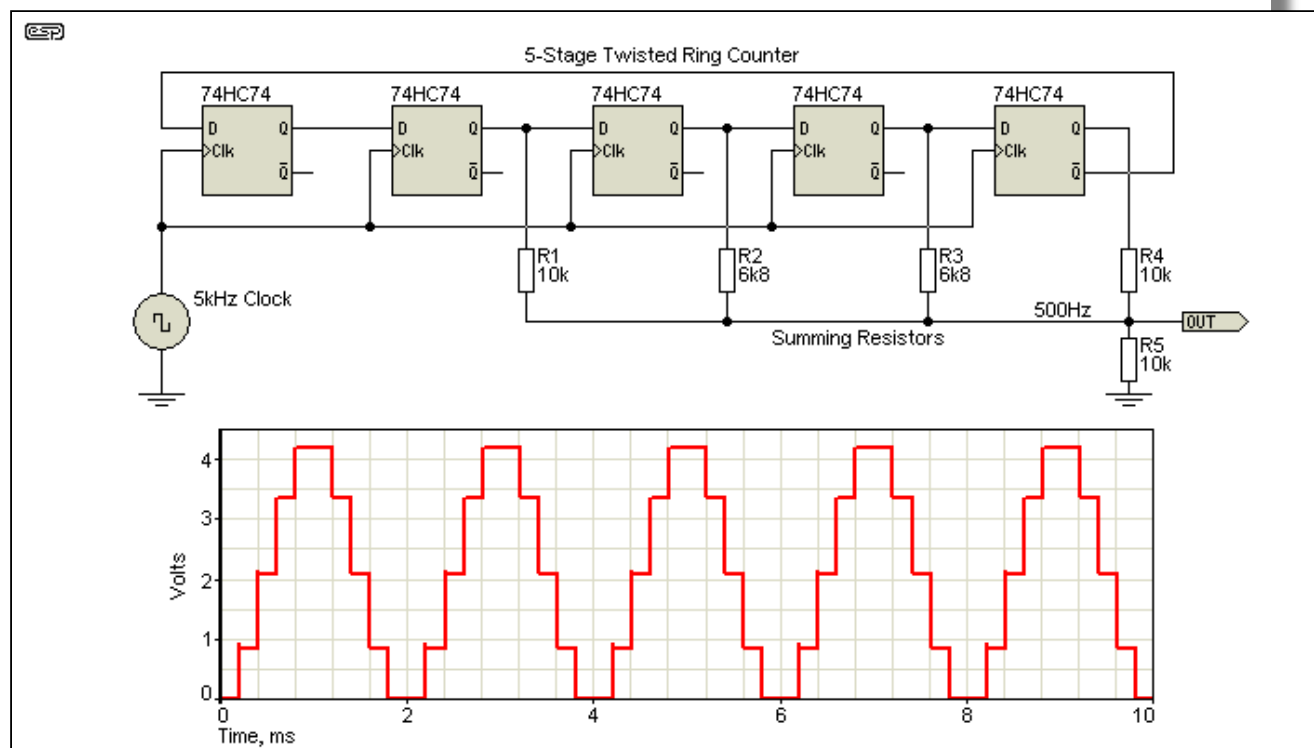


Figure 8.1 - Digital Sinewave Generation

The above is an example of a digitally generated sinewave. The output frequency is 1/10th of the input (clock) frequency. In the example shown it's limited to 5 bits, so while the *theoretical* distortion is 4%, this can only be achieved if the signal is filtered. More aggressive filtering will reduce the distortion. With a 220nF cap in parallel with R5, distortion is reduced to a little over 2%. This type of generator can only be used with a tracking filter as described in a Silicon Chip magazine article that had a design for a complete system.

